The Journal of Open Source Software

Limace.jl: A Julia package to compute hydromagnetic modes in spherical domains

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DOI: 10.xxxxxx/draft

Software

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Editor:

Submitted: 07 October 2024
Published: unpublished

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Summary

Hydromagnetic modes in spherical domains are relevant to the liquid cores of planets, moons or stars, as well as rotating fluid dynamics experiments. These modes are solutions to the linearized rotating magnetohydrodynamic equations that govern electrically conducting fluids under rapid rotation. `Limace.jl` is a package written in the Julia programming language (Bezanson et al., 2017), based on Galerkin projections of the governing equations onto trial vectors of the velocity and magnetic field. It aims to facilitate the calculation of modes in flexible setups with a high-level interface, whilst remaining computationally performant enough to tackle relevant physical parameters.

Statement of need

The study of hydromagnetic modes is relevant in particular to Earth's liquid core. Despite having been theoretically predicted a long time ago (Braginsky, 1970; Hide, 1966; Malkus, 1967), recent advances in numerical modelling and new observational evidence in geomagnetic data have reignited interest in these modes (Gerick et al., 2021; Gerick & Livermore, 2024; Gillet et al., 2022; Luo et al., 2022; Luo & Jackson, 2022; Triana et al., 2022). It is therefore relevant to the geophysical and astrophisical fluid dynamics community to have access to a code that models these mode. One of the only open-source models to compute hydromagnetic modes in relevant parameters for planetary cores is Kore (Triana et al., 2021), a spectral code based on ultraspherical polynomials in axisymmetric setups written in Python. The implementation of such a code requires substantial work due to the complexity of spectral equations that govern the modes. This is true also for very basic and idealized examples, resulting in a substantial entry barrier for scientists to model these modes. `Limace.jl` tries to lower this entry barrier, by providing an open source model with a very simple high-level API and modern online documentation with practical examples.

Despite having a high-level interface, `Limace.jl` can be used to solve complex and geophysically relevant problems. A unique feature of `Limace.jl` is the support of complex background magnetic fields and flows over which the modes evolve. The code has been developed from the beginning to leave assumptions of symmetry up to the user. The model code base is tested against mode solutions from the scientific literature to ensure its correctness.

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Theoretical background and implementation details

In non-linear form, the momentum equation of the incompressible fluid and the induction equation, are written as

$$\frac{\partial \mathbf{U}}{\partial t} + (\nabla \times \mathbf{U}) \times \mathbf{U} + 2\Omega \times \mathbf{U} = -\frac{1}{\rho}\nabla P + \frac{1}{\rho\mu_0}(\nabla \times \mathbf{B}) \times \mathbf{B} + \nu\nabla^2\mathbf{U} + \mathbf{F},$$
$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{U} \times \mathbf{B}) + \eta\nabla^2\mathbf{B},$$

with $\mathbf{U}$ the velocity, $\mathbf{B}$ the magnetic field, Ω the rotation axis, ρ the fluid density, P the reduced hydrodynamic pressure, μ_0 the magnetic permeability of free space, ν the kinematic viscosity, $\mathbf{F}$ some additional body force, and η the magnetic diffusivitiy.

In order to compute modal solutions, the velocity, magnetic and pressure fields a linearized, so that

$$\mathbf{U}(\mathbf{r},t) = \mathbf{U}_0(\mathbf{r}) + \mathbf{u}(\mathbf{r})e^{\lambda t},$$
$$\mathbf{B}(\mathbf{r},t) = \mathbf{B}_0(\mathbf{r}) + \mathbf{b}(\mathbf{r})e^{\lambda t},$$
$$P(\mathbf{r},t) = P_0(\mathbf{r}) + p(\mathbf{r})e^{\lambda t}.$$

with $\lambda = -\sigma + \mathrm{i}\omega$, with σ the damping rate and ω the frequency of the oscillatory perturbation to the steady background. Removing the steady part and neglecting higher order terms, the linearized momentum and induction equations then read

$$\begin{aligned}\lambda\mathbf{u} =& -(\nabla \times \mathbf{u}) \times \mathbf{U}_0 - (\nabla \times \mathbf{U}_0) \times \mathbf{u} - 2\Omega\mathbf{e}_z \times \mathbf{u} - \frac{1}{\rho}\nabla p \\ &+ \frac{1}{\rho\mu_0}((\nabla \times \mathbf{b}) \times \mathbf{B}_0 + (\nabla \times \mathbf{B}_0) \times \mathbf{b}) + \nu\nabla^2\mathbf{u}, \\ \lambda\mathbf{b} =& \nabla \times (\mathbf{U}_0 \times \mathbf{b}) + \nabla \times (\mathbf{u} \times \mathbf{B}_0) + \eta\nabla^2\mathbf{b}.\end{aligned}$$

These equations are then projected onto trial vectors $\mathbf{f}_i$, so that

$$a_{ij} = \int \mathbf{f}_i \cdot \mathbf{a}\left(\mathbf{u}_j, \mathbf{b}_j, \mathbf{U}_0, \mathbf{B}_0\right) \mathrm{d}V,$$

where $\mathbf{a}$ is any of the terms in the momentum and induction equation and $\mathbf{f}_i = [\mathbf{u}_i, \mathbf{b}_i]$. Due to the divergence free condition on the velocity and magnetic field, i.e. the flow is incompressible and no magnetic monopoles exist, it is convenient to decompose the fields into poloidal and toroidal components, so that

$$\mathbf{u} = \sum_i \alpha_i \mathbf{u}_i = \sum_{l,m,n} \alpha^P_{lmn}\mathbf{P}_{lmn} + \sum_{l,m,n} \alpha^Q_{lmn}\mathbf{Q}_{lmn},$$
$$\mathbf{b} = \sum_i \beta_i \mathbf{b}_i = \sum_{l,m,n} \beta^S_{lmn}\mathbf{S}_{lmn} + \sum_{l,m,n} \beta^T_{lmn}\mathbf{T}_{lmn},$$

with $\alpha_i, \beta_i \in \mathbb{C}$. The respective poloidal and toroidal basis vectors are

$$[\mathbf{P}, \mathbf{S}]_{lmn} = \nabla \times \nabla \times [P, S]_{ln}(r)Y_l^m(\theta,\phi)\mathbf{r},$$
$$[\mathbf{Q}, \mathbf{T}]_{lmn} = \nabla \times [Q, T]_{ln}(r)Y_l^m(\theta,\phi)\mathbf{r}.$$

Here, $Y_l^m(\theta,\phi)$ is the (fully normalized) spherical harmonic of degree l and order m. The boundary conditions (or regularity condition at $r = 0$) are imposed on the scalar functions P, S, Q, T. The scalar functions can be chosen to have optimal properties, i.e. the resulting basis is orthogonal w.r.t a given inner product (Chen et al., 2018; Gerick & Livermore, 2024; Livermore, 2010). `Limace.jl` provides several optimal bases that satisfy relevant boundary conditions.

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We need to consider all combinations of poloidal and toroidal vector combinations in the projection of the forces. This leads to several long coupling terms, especially for the Lorentz force, advection and induction terms. The integrals of these coupling terms over the spherical surfaces are computed through the Adam-Gaunt and Elsasser variables (James, 1973), which are calculated from Wigner symbols (available in Julia through WignerSymbols.jl, based on Johansson & Forssén (2016)). The remaining integration in radial direction is done using Gauss-Legendre quadratures, available through FastGaussQuadrature.jl. The exact modelled equations are outlined in Gerick & Livermore (2024), based on the work of Ivers & Phillips (2008).

From the projected equations, the problem reduces to a generalized eigen problem

$$\lambda \mathbf{A}\mathbf{x} = \mathbf{B}\mathbf{x},$$

that is solved numerically. Here, the eigenvector $\mathbf{x}$ contains the coefficients α_i (and β_i). The matrix $\mathbf{B}$ is generally not symmetric/Hermitian, but $\mathbf{A}$ can be the unit matrix, symmetric tridiagonal or symmetric, depending on the chosen bases.

For small problem sizes, the eigen problem can be solved using dense methods, e.g. using the standard library function `LinearAlgebra.eigen`. To compute few eigen solutions of the sparse system, a shift-invert spectral transform method is provided, based on the sparse LU factorization from `UMFPACK` (Davis, 2004) and the partial Schur decomposition implemented in ArnoldiMethod.jl (Stoppels & Nyman, n.d.).

For postprocessing, `Limace.jl` uses a fast spherical harmonic transform implemented in the SHTns library (Schaeffer, 2013), and available in Julia through SHTns.jl. It is used to transform the spectral coefficients to vector fields evaluated on a spatial grid. `Limace.jl` does not provide any plotting routines, but some examples are given, leaving the choice of plotting library up to the user.

Basic Example - Malkus background magnetic field

In this example, we calculate the spectrum of modes when the background field is $\mathbf{B}_0 = s\mathbf{e}_\phi$ (Malkus, 1967).

```
using Limace, LinearAlgebra

N = 6
u = Inviscid(N)
b = PerfectlyConducting(N) # == Inviscid(N)

B0 = BasisElement(b, Toroidal, (1,0,0), 2sqrt(2pi/15)) # corresponds to B_0 = s e_phi
Le = 1e-2

RHSc = Limace.coriolis(u)
RHSl = Limace.lorentz(u, b, B0)
RHSi = Limace.induction(b,u,B0)
RHSd = spzeros(length(b),length(b)) #empty (no Ohmic diffusion)
RHS = [RHSc/Le RHSl
       RHSi RHSd];

λ = eigvals(Matrix(RHS))
```

In this simple configuration analytical solutions can be derived (Malkus, 1967), relating the frequency of the hydromagnetic problem to the inertial mode frequencies (Zhang et al., 2001). We verify that our calculated mode spectrum contains some of the analytical solutions:

```julia
function zhang(m, N)
    sm = sign(m)
    m = abs(m)
    return -sm*2 / (m + 2) * (√(1 + m * (m + 2) / (N * (2N + 2m + 1))) - 1) * im
end

slow(m, N, Le, λ = imag(zhang(m, N))) = im * λ / 2Le * (1 - √(1 + 4Le^2 * m * (m - λ) /
fast(m, N, Le, λ = imag(zhang(m, N))) = im * λ / 2Le * (1 + √(1 + 4Le^2 * m * (m - λ) /

using Test

@testset "Malkus modes" begin
    for m = vcat(-(N-1):-1, 1:(N-1))
        @test any(isapprox(slow(m,1,Le)),λ)
        @test any(isapprox(fast(m, 1, Le)),λ)
    end
end
```

More examples are given in the documentation of `Limace.jl`.

Outlook

The code is written to be extendable, both in terms of the chosen Galerkin bases and also additional forcings. In the future, the code can be extended to fully support spherical shells and a heat equation, following the spectral equations in Ivers & Phillips (2008).

Acknowledgements

I have received funding from the European Research Council (ERC) GRACEFUL Synergy Grant No. 855677. This project has been funded by ESA in the framework of EO Science for Society, through contract 4000127193/19/NL/IA (SWARM + 4D Deep Earth: Core).

I thank Phil Livermore for the key contributions in the theoretical development of the model.

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